

Practical 5

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The screenshot displays the SQL Enterprise Manager interface. On the left, the 'Navigator' pane shows the 'MANAGEMENT' section with options like 'Server Status', 'Client Connections', 'Users and Privileges', 'Status and System Variables', 'Data Export', and 'Data Import/Restore'. Below this is the 'INSTANCE' section with 'Startup / Shutdown', 'Server Logs', and 'Options File'. The 'PERFORMANCE' section has 'Administration' and 'Schemas' tabs. The 'Information' pane at the bottom shows 'No object selected'.

The main window, titled 'Query 1 library_db', shows a SQL query editor with the following text:

```
1 CREATE DATABASE library_db;
2
3
```

Below the query editor, the 'Output' pane shows the execution results. It includes a table with columns for '#', 'Time', 'Action', and 'Message'.

#	Time	Action	Message
7	20:05:53	SELECT * FROM students LIMIT 0, 50000	3 row(s) returned
8	09:38:32	CREATE DATABASE library_db	1 row(s) affected

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Administration
- Schemas

Information

No object selected

Query 1 library_db

```
1 CREATE DATABASE library_db;
2 CREATE TABLE IF NOT EXISTS members (
3     id INT AUTO_INCREMENT PRIMARY KEY,
4     username VARCHAR(100) UNIQUE NOT NULL,
5     password_hash VARCHAR(255) NOT NULL,
6     role VARCHAR(30) NOT NULL DEFAULT 'member'
7 );
8
9 CREATE TABLE IF NOT EXISTS books (
10    id INT AUTO_INCREMENT PRIMARY KEY,
11    title VARCHAR(250) NOT NULL,
12    author VARCHAR(200) NOT NULL,
13    isbn VARCHAR(13) UNIQUE NOT NULL,
14    price DECIMAL(10,2) NOT NULL,
15    published_date DATE NOT NULL,
16    description TEXT,
17    embedding JSON
18 );
19
20
21
```

Output

Action Output

#	Time	Action	Message
8	09:38:32	CREATE DATABASE library_db	1 row(s) affected
9	09:38:53	CREATE DATABASE library_db	Error Code: 1007. Can't create database 'library_db'; database exists
10	09:40:58	CREATE TABLE IF NOT EXISTS members (id INT AUTO_INCREMENT PRIMARY KEY, username VAR...	0 row(s) affected

Object Info Session

File Edit View Query Database Server Tools Scripting Help

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No object selected

Query 1 library_db

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```

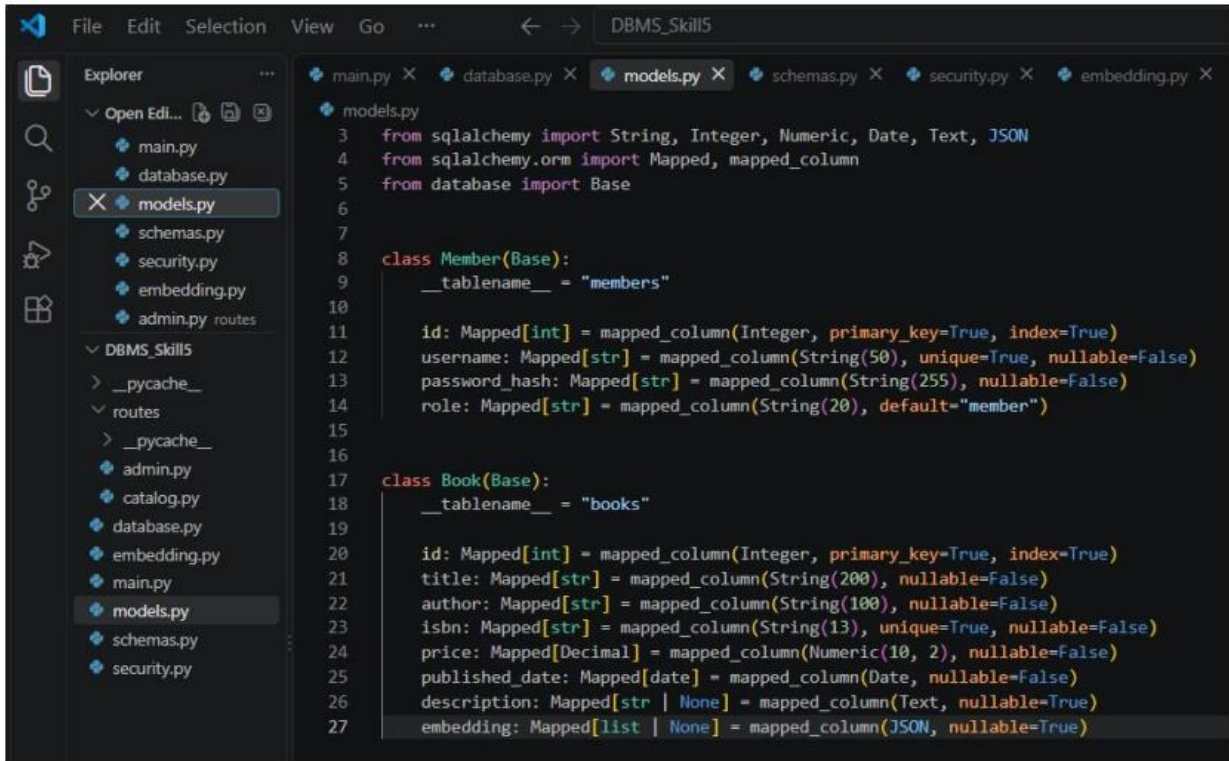
Output

Action Output

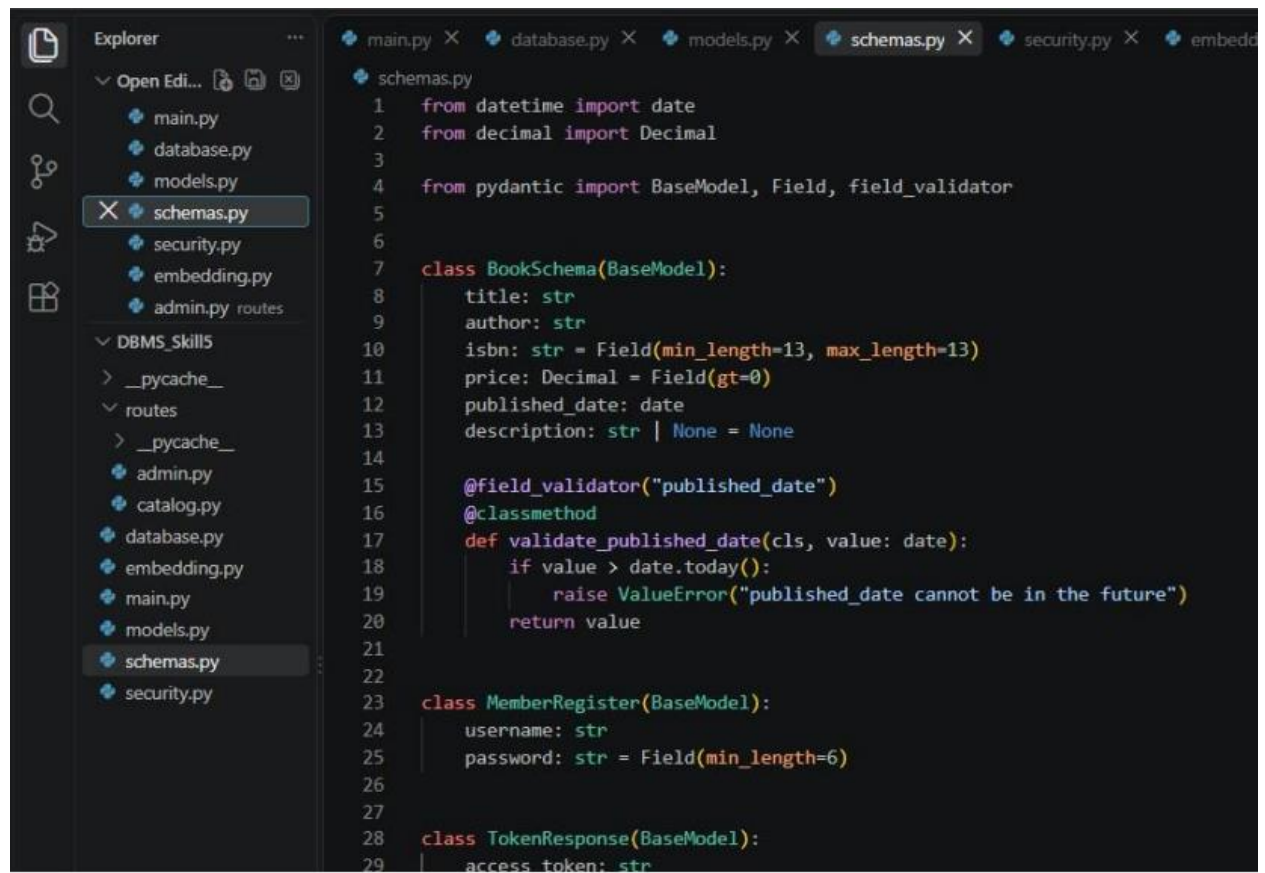
#	Time	Action	Message
9	09:38:53	CREATE DATABASE library_db	Error Code: 1007. Can't create database 'library_db'; database exists
10	09:40:58	CREATE TABLE IF NOT EXISTS members (id INT AUTO_INCREMENT PRIMARY KEY, username VAR...	0 row(s) affected
11	09:41:27	CREATE TABLE IF NOT EXISTS books (id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(2...	0 row(s) affected

Object Info Session

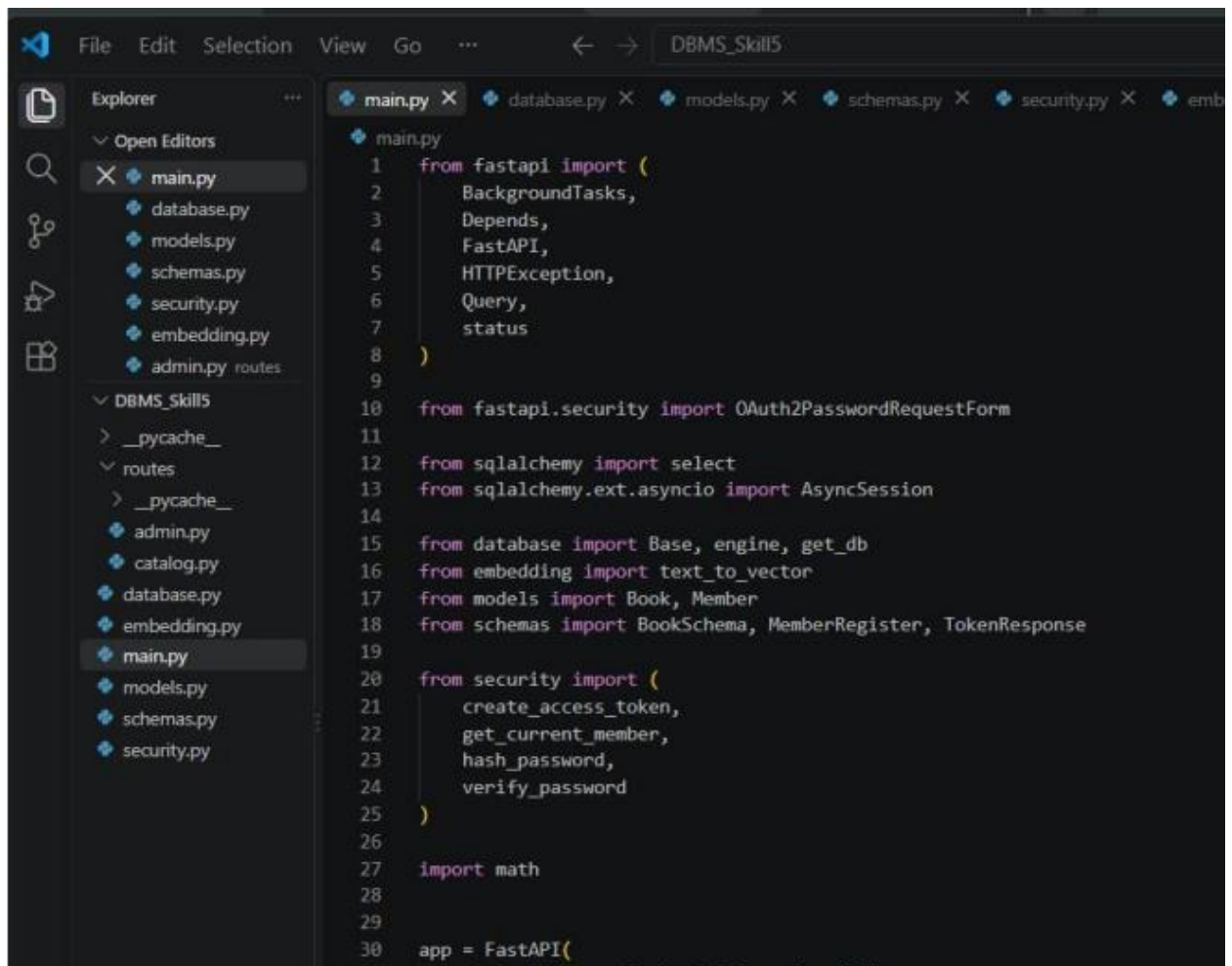
```
PS C:\library_api> python -m pip install fastapi uvicorn
>> python -m pip install "python-jose[cryptography]"
>> python -m pip install passlib[bcrypt]
● >> python -m pip install python-multipart
```

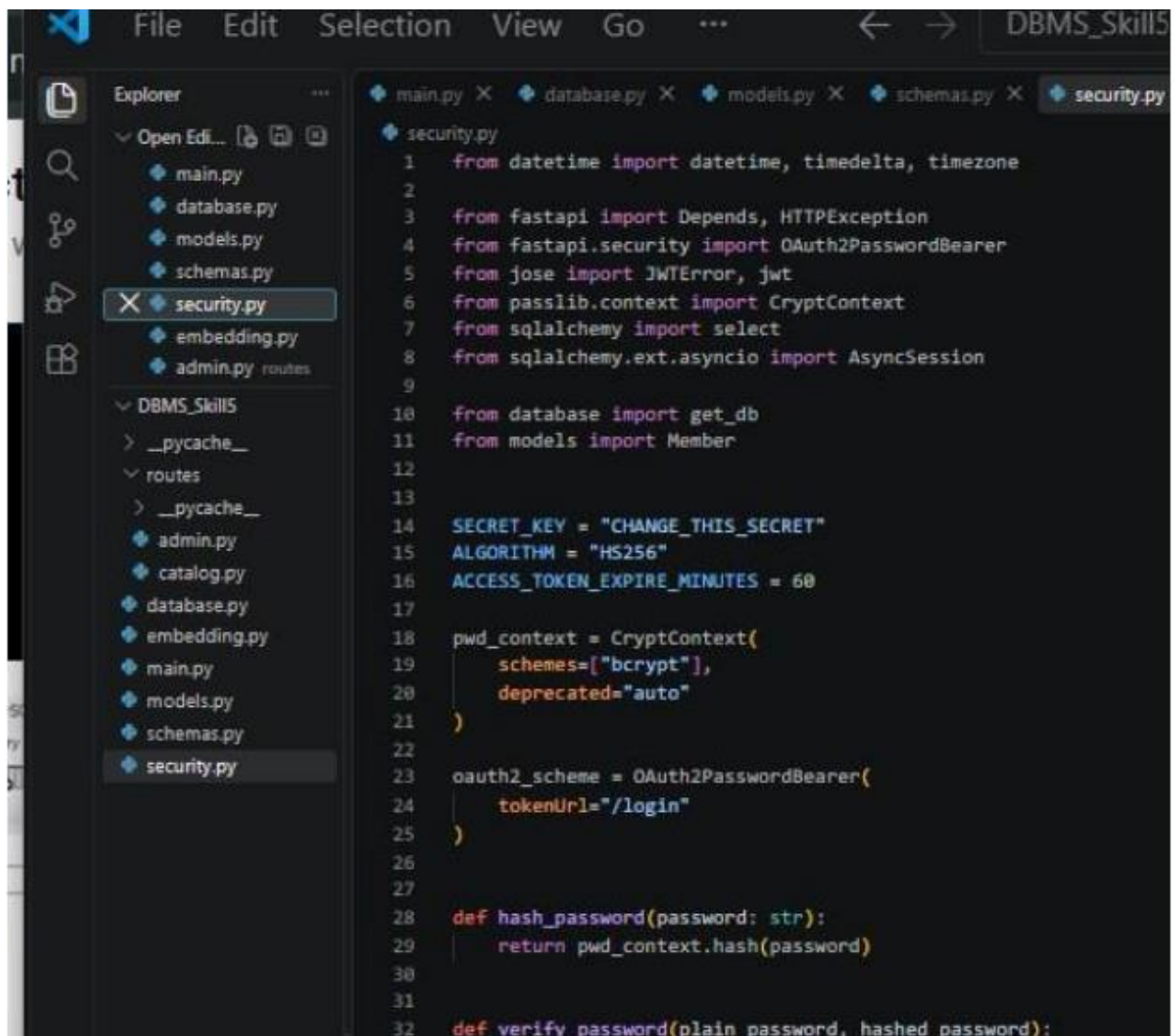


```
models.py
3 from sqlalchemy import String, Integer, Numeric, Date, Text, JSON
4 from sqlalchemy.orm import Mapped, mapped_column
5 from database import Base
6
7
8 class Member(Base):
9     __tablename__ = "members"
10
11     id: Mapped[int] = mapped_column(Integer, primary_key=True, index=True)
12     username: Mapped[str] = mapped_column(String(50), unique=True, nullable=False)
13     password_hash: Mapped[str] = mapped_column(String(255), nullable=False)
14     role: Mapped[str] = mapped_column(String(20), default="member")
15
16
17 class Book(Base):
18     __tablename__ = "books"
19
20     id: Mapped[int] = mapped_column(Integer, primary_key=True, index=True)
21     title: Mapped[str] = mapped_column(String(200), nullable=False)
22     author: Mapped[str] = mapped_column(String(100), nullable=False)
23     isbn: Mapped[str] = mapped_column(String(13), unique=True, nullable=False)
24     price: Mapped[Decimal] = mapped_column(Numeric(10, 2), nullable=False)
25     published_date: Mapped[date] = mapped_column(Date, nullable=False)
26     description: Mapped[str | None] = mapped_column(Text, nullable=True)
27     embedding: Mapped[list | None] = mapped_column(JSON, nullable=True)
```



```
3
4 VECTOR_DIM = 384
5
6
7 def text_to_vector(text: str) -> list[float]:
8     values = []
9
10    for i in range(VECTOR_DIM):
11        digest = hashlib.sha256(
12            f"{text.lower():{i}".encode()
13        ).digest()
14
15        number = int.from_bytes(
16            digest[:4],
17            byteorder="big"
18        )
19
20        values.append(
21            (number / 2**32) * 2 - 1
22        )
23
24    magnitude = math.sqrt(
25        sum(x * x for x in values)
26    )
27
```





The image shows a code editor window with a dark theme. The top menu bar includes File, Edit, Selection, View, Go, and navigation arrows. The title bar on the right says "DBMS_Skills5". The Explorer sidebar on the left shows a project structure with files like main.py, database.py, models.py, schemas.py, security.py, embedding.py, and admin.py. The security.py file is selected and open in the editor. The code in security.py includes imports for datetime, timedelta, timezone, fastapi dependencies, JWT, passlib, sqlalchemy, and database models. It defines a SECRET_KEY, ALGORITHM, and ACCESS_TOKEN_EXPIRE_MINUTES. It also initializes a CryptContext and an OAuth2PasswordBearer. Finally, it defines two functions: hash_password and verify_password.

```
1 from datetime import datetime, timedelta, timezone
2
3 from fastapi import Depends, HTTPException
4 from fastapi.security import OAuth2PasswordBearer
5 from jose import JWTError, jwt
6 from passlib.context import CryptContext
7 from sqlalchemy import select
8 from sqlalchemy.ext.asyncio import AsyncSession
9
10 from database import get_db
11 from models import Member
12
13
14 SECRET_KEY = "CHANGE_THIS_SECRET"
15 ALGORITHM = "HS256"
16 ACCESS_TOKEN_EXPIRE_MINUTES = 60
17
18 pwd_context = CryptContext(
19     schemes=["bcrypt"],
20     deprecated="auto"
21 )
22
23 oauth2_scheme = OAuth2PasswordBearer(
24     tokenUrl="/login"
25 )
26
27
28 def hash_password(password: str):
29     return pwd_context.hash(password)
30
31
32 def verify_password(plain_password, hashed_password):
```

